

CS101: Introduction to C Programming Mid-Sem Lab Exam: (Set I)

Duration: 2 hours (8:35 AM to 10:35 AM)

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**DEPARTMENT OF COMPUTER SCIENCE ENGINEERING
INDIAN INSTITUTE OF TECHNOLOGY DHARWAD**

October 1, 2024

Lab Instructions

Please abide by the following instructions for the entire duration of the semester.

1. You are restricted from connecting to the Internet or using mobile phones during this lab exam.
2. You will be provided rough papers to work out if required.
3. **Any malpractice observed will be reported and disciplinary action will be taken.**

Before you start programming:

1. In the `home` directory, you can see a directory named `Mini Lab 2.zip`. Right-click and select **Extract Here** to extract the `.zip` file. Now open the folder `Mini Lab 2` which contains the question paper, and an empty sub-directory named *Submissions*.
2. You need to write all your programs in this *Submissions* directory.
3. **Note:** The *Submissions* directory should contain the following files:
`roll_number_1.c` for question 1
`roll_number_2.c` for question 2
`roll_number_3.c` for question 3
`roll_number_4.c` for question 4
Here, `roll_number` is your institute roll number. Follow these naming conventions otherwise your programs will not be evaluated.

After you complete programming and before leaving the lab

1. Ensure your assigned TA has marked your attendance.
2. Make sure not to delete the program files.

Note: All questions are compulsory. Students are expected to complete the lab exam in 2 hours.

1. (3 marks) Create a file named `roll_number_1.c` in the *Submissions* directory.

Write a C program that prompts the user to take two positive integer inputs namely, rows `r` and columns `c`. Prompt the user to enter integer inputs for the matrix elements. Pass these two matrices to a function that calculates the sum of these two matrices and prints the resultant matrix in the function itself.

NOTE 1: The rows and columns should be positive integers and other input should be printed as invalid.

NOTE 2: The array elements should be of type `int` if it is a character then print invalid.

Sample Input 1

Enter number of rows and columns: 2 2

Enter elements of the first matrix:

1 2

3 4

Enter elements of the second matrix:

5 6

7 8

Sample Output 1

Resulting Matrix after addition:

6 8

10 12

Sample Input 2

Enter the number of rows and columns: 2 -2

Sample Output 2

Invalid matrix size. Both rows and columns must be positive integers

Sample Input 3

Enter number of rows and columns: 1 1

Enter elements of the first matrix:

10

Enter elements of the second matrix:

5

Sample Output 3

Resulting Matrix after addition:

15

Sample Input 4

Enter number of rows and columns: 2 2

Enter elements of the first matrix:

8 a

6 5

Enter elements of the second matrix:

4 3

2 1

Sample Output 4

Invalid input. Please enter valid integers for all matrix elements.

2. (5 marks) Create a file named `roll_number_2.c` in the *Submissions* directory.

Write a C program that prompts the user to take an integer input n , the dimension of a square matrix ($n * n$), and input integer elements into the square matrix. Pass the matrix to a function that checks whether the given matrix is *symmetric*.

Definition: Symmetric Matrix is a square matrix that is equal to its transpose, meaning that the elements of the matrix are symmetric about the main diagonal.

NOTE 1: n should be a positive integer and any other input should be printed as invalid.

NOTE 2: The array elements should be of type `int` if it is a character then print invalid.

Sample Input 1

Enter size of square matrix (n x n): 2

Enter elements of matrix:

1 2

2 1

Sample Output 1

The matrix is symmetric.

Sample Input 2

Enter size of square matrix (n x n): 3

Enter elements of matrix:

1 2 3

2 5 6

4 6 9

Sample Output 2

The matrix is not symmetric.

Sample Input 3

Enter size of square matrix (n x n): -3

Sample Output 3

Invalid matrix size. The size of the square matrix (n x n) must be a positive integer.

Sample Input 4

Enter size of square matrix (n x n): 1

Enter elements of matrix:

42

Sample Output 4

The matrix is symmetric.

Sample Input 5

Enter size of square matrix (n x n): 3

Enter elements of matrix:

1 2 a

4 5 6

7 8 9

Sample Output 5

Invalid input. Please enter valid integers for all matrix elements.

3. (5 marks) Create a file named `roll_number_3.c` in the *Submissions* directory.

Write a C program that prompts the user to enter *two* strings. Now, concatenate these two strings within a new character array without using any built-in string methods and print it in the main function.

Definition: String concatenation is to combine two or more strings and store them in a single string.

Sample Input 1

Enter first string: @Hello!

Enter second string: #World123

Sample Output 1

Concatenated string: @Hello!#World123

Sample Input 2

Enter first string: fine

Enter second string: World

Sample Output 2

Concatenated string: fineWorld

Sample Input 3

Enter first string: HELLO

Enter second string: world

Sample Output 3

Concatenated string: HELLOworld

4. (7 marks) Create a file named `roll_number_4.c` in the *Submissions* directory.
Write a C program that prompts users to input a word/sentence in each row of a 2D character array. Pass the array to a function `replaceVowels(char strings[10][100], int numStrings)` that replaces vowels in each string with the character '*' and prints the modified strings inside the function itself.

Sample Input 1

Enter number of strings: 2

Enter string 1: Hello

Enter string 2: World

Sample Output 1

Modified string 1: H*ll*

Modified string 2: W*rld

Sample Input 2

Enter number of strings: 3

Enter string 1: Bcd

Enter string 2: Fly

Enter string 3: Zzzz

Sample Output 2

Modified string 1: Bcd

Modified string 2: Fly

Modified string 3: Zzzz

Sample Input 3

Enter number of strings: 1

Enter string 1: aeiouAEIOU

Sample Output 3

Modified string 1: *****

Sample Input 4

Enter number of strings: 2

Enter string 1: The sky is blue.

Enter string 2: 12345 Aeiou!

Sample Output 4

Modified string 1: Th* sky *s bl**.

Modified string 2: 12345 *****!

CS101: Introduction to C Programming Mid-Sem Lab Exam: (Set II)

Duration: 2 hours (8:35 AM to 10:35 AM)

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Here, `roll_number` is your institute roll number. Follow these naming conventions otherwise your programs will not be evaluated.

After you complete programming and before leaving the lab

1. Ensure your assigned TA has marked your attendance.
2. Make sure not to delete the program files.

Note: All questions are compulsory. Students are expected to complete the lab exam in 2 hours.

1. (3 marks) Create a file named `roll_number_1.c` in the *Submissions* directory.

Write a C program that prompts the user to take two positive integer inputs for rows `r` and columns `c`. Prompt the user to enter integer inputs for the matrix elements. Pass these two matrices to a function that subtracts these two matrices and prints the resultant matrix in the function itself.

NOTE 1: The rows and columns should be positive integers and other input should be printed as invalid.

NOTE 2: The array elements should be of type `int` if it is a character then print invalid.

Sample Input 1

Enter number of rows and columns: 2 2

Enter elements of the first matrix:

8 7

6 5

Enter elements of the second matrix:

4 3

2 1

Sample Output 1

Resulting Matrix after subtraction:

4 4

4 4

Sample Input 2

Enter the number of rows and columns: 2 -2

Sample Output 2

Invalid matrix size. Both rows and columns must be positive integers

Sample Input 3

Enter number of rows and columns: 1 1

Enter elements of the first matrix:

10

Enter elements of the second matrix:

5

Sample Output 3

Resulting Matrix after subtraction:

5

Sample Input 4

Enter number of rows and columns: 2 2

Enter elements of the first matrix:

8 a

6 5

Enter elements of the second matrix:

4 3

2 1

Sample Output 4

Invalid input. Please enter valid integers for all matrix elements.

2. (5 marks) Create a file named `roll_number_2.c` in the *Submissions* directory.

Write a C program that prompts the user to take an integer input n , the dimension of a square matrix ($n * n$), and input integer elements into the square matrix. Pass the matrix to a function that prints the lower triangular matrix and then the upper triangular matrix for the given matrix.

Definitions

- (a) Lower triangular matrix: A matrix with all elements above the main diagonal equal to zero.
- (b) Upper triangular matrix: A matrix with all elements below the main diagonal equal to zero.

NOTE 1: n should be a positive integer and any other input should be printed as invalid.

NOTE 2: The array elements should be of type `int` if it is a character then print invalid.

Sample Input 1

Enter size of square matrix (n x n): 3

Enter elements of matrix:

1 2 3

4 5 6

7 8 9

Sample Output 1

Lower Triangular Matrix:

1 0 0

4 5 0

7 8 9

Upper Triangular Matrix:

1 2 3

0 5 6

0 0 9

Sample Input 2

Enter size of square matrix (n x n): -3

Sample Output 2

Invalid matrix size. The size of the square matrix (n x n) must be a positive integer.

Sample Input 3

Enter Enter size of square matrix (n x n): 1

Enter elements of matrix:

42

Sample Output 3

Lower Triangular Matrix:

42

Upper Triangular Matrix:

42

Sample Input 4

Enter size of square matrix (n x n): 3

Enter elements of matrix:

0 0 0

0 0 0

0 0 0

Sample Output 4

Lower Triangular Matrix:

0 0 0

0 0 0

0 0 0

Upper Triangular Matrix:

0 0 0

0 0 0

0 0 0

3. (5 marks) Create a file named `roll_number_3.c` in the *Submissions* directory.

Write a C program that prompts the user to enter *two* strings. Implement a logic to compare these two strings without using any built-in string methods.

NOTE 1: the strings are **case-sensitive**, meaning 'a' is not same as 'A'.

NOTE 2: The strings should not be empty.

Sample Input 1

Enter first string: apple

Enter second string: Apple

Sample Output 1

The strings are not same

Sample Input 2

Enter first string: name

Enter second string: name

Sample Output 2

The strings are same

Sample Input 3

Enter first string: 1234

Enter second string: 1234

Sample Output 3

The strings are same

4. (7 marks) Create a file named `roll_number_4.c` in the *Submissions* directory.

Develop a C program that prompts the user to input an integer variable n , representing the number of strings, and subsequently requests the user to enter a word or sentence for each row of a 2D character array. The program should then pass this array to a function:

```
void replaceConsonants(char strings[n][100], int numStrings)
```

which will replace all consonants in each string with the character `+` and print the modified strings within the function itself.

NOTE: The strings should not be empty

Sample Input 1

Enter number of strings: 2

Enter string 1: Hello

Enter string 2: World

Sample Output 1

Modified string 1: +e++o

Modified string 2: +o++++

Sample Input 2

Enter number of strings: 2

Enter string 1: AeIoU

Enter string 2: bCdEf

Sample Output 2

Modified string 1: AeIoU

Modified string 2: +++E+

Sample Input 3

Enter number of strings: 1

Enter string 1: aeiouAEIOU

Sample Output 3

Modified string 1: aeiouAEIOU

Sample Input 4

Enter number of strings: 1

Enter string 1: Programming is fun!

Sample Output 4

Modified string 1: ++o++a++i++ i+ +u+!